

Contact

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www.linkedin.com/in/rafaelcp
(LinkedIn)

webacademico.canoas.ifrs.edu.br/
~rcpinto/ (Personal)

Top Skills

Artificial Intelligence

Machine Learning

Evolutionary Computation

Languages

Spanish (Elementary)

French (Elementary)

English (Professional Working)

Portuguese (Native or Bilingual)

Japanese (Elementary)

Certifications

Ciências da Vida na Escola:
Neurociências Aplicada à Educação
(CVE)

Natural Language Processing with
Classification and Vector Spaces

3rd Neuro-Symbolic AI Summer
School 2024

Natural Language Processing with
Attention Models

Deep Learning Specialization

Honors-Awards

Patron of the 2016' IFRS Canoas
Computer Technician Course

Japanese Language Oratory Contest
- 3rd Place at 'D' Category

Patron of the 2017' IFRS Canoas
Technologist Course in Software
Analysis and Development

Patron of the 2018' IFRS Canoas
Technologist Course in Software
Analysis and Development

Honoured Teacher - 2019' IFRS
Canoas Technologist Course in
Software Analysis and Development

Rafael Pinto

AI Professor at IFRS

Greater Porto Alegre

Summary

Started in 1998 as a desktop programmer with Delphi and Visual Basic. Web developer from 2002 to 2013. Currently, a professor at IFRS - Canoas. PhD in Machine Learning.

My passion is Artificial Intelligence. In my undergrad thesis, I developed a neocortex inspired algorithm using a hierarchy of Kohonen Self-Organizing Maps, based on the Memory-Prediction Framework. In my master's thesis, I worked with recurrent neural networks and integrated them with Gaussian mixture models. I've also worked on combining reinforcement learning and incremental Gaussian mixture models for my Ph.D. dissertation. My goals are to teach computer science (AI) (done!) and to create advanced AI.

Also, more as a hobby, I have developed a web-based massive game with PHP and AJAX (back when AJAX was new) and some smaller Flash and HTML5 games. I'm also working on an 8-bit Mega Man fan game.

Specialties:

Computer Science: Neural Networks, Gaussian Mixture Models, Reinforcement Learning, Evolutionary Computing and Machine Learning in general.

Technology: I have strong expertise and experience with PHP, JavaScript, HTML, CSS, 2D Game Development, Matlab, Python, Pytorch.

Experience

IFRS

Professor

February 2013 - Present (13 years 8 months)

Publications

LoopSOM: a robust som variant using self-organizing temporal feedback connections

Fast Reinforcement Learning with Incremental Gaussian Mixture Models

Neuroevolution of Self-Attention Over Proto-Objects

Echo State Incremental Gaussian Mixture Network for Spatio-Temporal Pattern Processing

One-Shot Learning in the Road Sign Problem

Teaching and researching.

Federal University of Rio Grande do Sul

Postdoctoral Researcher

July 2024 - July 2025 (1 year 1 month)

Porto Alegre, Rio Grande do Sul, Brazil

Ministério Público Federal

Software Developer

June 2006 - February 2013 (6 years 9 months)

Software development with PHP, HTML, CSS, JavaScript, AJAX, Pyke, Zope, Java, Python, MySQL and Oracle.

CEUE

Volunteer Math Teacher

February 2011 - March 2011 (2 months)

Teaching.

Procempa

Intern Web Developer

2004 - 2006 (2 years)

Web developer with PHP, HTML, CSS, Java Script and MySQL. Maintained big content portals for local government, ensuring accessibility.

Eicomm

Intern Software Developer

2002 - 2004 (2 years)

Web developer with ASP, HTML, CSS and Java Script, using Firebird and Access databases. Desktop programmer with Delphi. Worked on image processing software.

MGR Informática

Programmer

2001 - 2001 (less than a year)

School management software programmer with Visual Basic and SQL Server.

Flex Informática

Programmer

1999 - 2000 (1 year)

Desktop programmer with Delphi.

Education

Federal University of Rio Grande do Sul

Ph.D., Neural Networks and Reinforcement Learning · (2012 - 2017)

Universidade Federal do Rio Grande do Sul

MSc, Artificial Intelligence · (2009 - 2011)

Federal University of Rio Grande do Sul

BSc, Computer Science · (2002 - 2009)

Centro Universitário Claretiano

Licentiate degree · (2014 - 2014)

NELE

Japanese Language · (2006 - 2007)